

Two Honest Nulls on the WANDERING Mechanism Hypothesis: Pre-registered Causal Tests Refute Simple Mid-Layer-to-Edge Rescue

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Abstract

Companion paper to “Tool-Entropy Collapse: A Cross-Architecture Signature of Agent WANDERING Failure”. The Tool-Entropy paper proposed a mechanism hypothesis (§13 therein) for the WANDERING failure mode: mid-layer verdict consolidated (“model knows it is done”) but edge circuits fail to translate verdict into `finish_tool` action emission. We pre-register two causal tests of this hypothesis on the same Qwen3.6-27B SWE-bench Pro dataset (N=99 trajectories at L11/L23/L31/L43/L55).

Experiment D (Forced-Finish Counterfactual) asks whether WANDERING agent-patches pass tests at the SUCCESS rate, which would support “WANDERING knew the answer”. Across 89 Docker-evaluated instances, WANDERING agent-patch pass rate is 21.1% (4/19), SUCCESS is 15.4% (6/39), odds ratio 1.47, Fisher exact $p = 0.71$. **Null:** patch quality does not distinguish sub-classes.

Experiment B (L55 SUCCESS-Donor Steering) asks whether persistent injection of a SUCCESS-derived L55 residual biases WANDERING agents toward `finish_tool` emission. Twenty WANDERING trajectories were re-run on H100 with and without an always-on `mode=add`, $\alpha = 0.3$ hook. Primary (with hook): 6/20 (30%); Baseline (no hook): 7/20 (35%); Fisher $p = 1.00$; odds ratio 0.796 (slight HURT direction); McNemar (3 baseline-only flips vs 2 hook-only) not significant. **Null with reverse hint:** SUCCESS-direction L55 injection at $\alpha = 0.3$ is not a causal lever.

Methodological discovery: the Phase 6 WANDERING classification is not hardware-stable. The Tool-Entropy paper classified all 20 instances as `max_turns` (0/20 `finish_tool`) on an RTX 6000 Ada, but H100 with the same deterministic seed and no hook flips 35% (7/20) to `finish_tool`. Single-run cross-GPU classification captures a temperature-stochastic phenotype that does not transfer cleanly between hardware.

The Tool-Entropy mechanism hypothesis as stated is not rescued by simple residual perturbation. The empirical pattern joins *Two Forms of Epiphenomenal Probes* (Phase 7/8) as a third null result: the v5 tool-entropy signal detects WANDERING (correlatively) but the SUCCESS L55 direction at $\alpha = 0.3$ does not lever it (causally). Constructive next steps include α -sweep, cross-layer Plan B, transient surgical intervention at lock-in turn, and multi-seed WANDERING classification protocol for hardware-stable labels.

1 Background and Hypothesis Under Test

The Tool-Entropy Collapse paper [2] identifies WANDERING as a distinct sub-class of agent failure that is externally identical to LOCKED (both exhaust the turn budget without emitting `finish_tool`) but differs internally: WANDERING agents have probe scores > 0.70 throughout,

yet fail to translate this internal “confidence” into a terminating action. Tool-Entropy §13 proposes a mechanism:

WANDERING agents have consolidated a mid-layer verdict (the model “knows” it is done), but edge circuits (L11 surface processing, L55 output planning) that translate verdict $\rightarrow$ finish_tool action have not aligned. Decision-to-action circuit desynchronized.

Two empirical predictions follow:

1. **Patch-quality prediction (Exp D):** if WANDERING agents have “knowledge of done”, their generated patches (which were typed in the trajectory but not submitted via `finish_tool`) should pass SWE-bench Pro tests at a rate comparable to SUCCESS agents who did submit. The hypothesis predicts $\text{WANDERING-pass-rate} \geq \text{SUCCESS-pass-rate}$.
2. **Direction-injection prediction (Exp B):** if mid-layer verdict is consolidated and edge-layer alignment is the bottleneck, then injecting the SUCCESS-typical L55 residual (representing “what an aligned edge layer produces”) into a WANDERING trajectory’s forward pass should bias the agent toward `finish_tool` emission. The hypothesis predicts $\text{hooked-flip-rate} > \text{no-hook-baseline}$.

This paper pre-registers and runs both tests. Both are null.

2 Related Work

Pre-registered causal tests in mech interp. The closest methodological precedent is the “Two Forms of Epiphenomenal Probes” line [3], which formalized softmax-temperature and template-locked epiphenomena via structural-rigidity α -sweep diagnostic. This paper extends the same epistemic discipline: predict, intervene, falsify.

Activation patching as causal test. Standard cross-layer patching [5, 6] requires a clean and corrupted run and substitutes activations between them. Our Plan B departs from this by installing an always-on hook from turn 0 rather than performing a single-token surgical patch, sidestepping the determinism risk of trajectory replay (see §3).

SLDS-style continuous-dynamics interpretation. Recent work [1] argues residual streams behave more like continuous trajectories than discrete-phase processes. Our Exp B null is independently consistent with this: a single edge-layer residual direction does not partition the action space.

Failure-mode rescue interventions. Steering-based rescue of failure modes [7, 4] typically demonstrates positive direction with single-shot intervention. Plan B tests whether a persistent rescue intervention scales, and finds it does not for the WANDERING mode.

3 Methodology

3.1 Dataset and Model

We re-use the Phase 6 dataset from the Tool-Entropy paper: 99 SWE-bench Pro trajectories from Qwen3.6-27B with per-turn residuals captured at L11/L23/L31/L43/L55, classified into SUCCESS ($n = 40$), LOCKED ($n = 39$), and WANDERING ($n = 20$). All experiments here are restricted to the 20 WANDERING trajectories plus matched SUCCESS controls for donor extraction.

3.2 Experiment D: Forced-Finish Counterfactual

We run the SWE-bench Pro official Docker evaluation pipeline on each trajectory’s submitted-but-not-finalized patch across three conditions:

- **none**: no patch applied (baseline, expected $\sim 0\%$ pass)
- **golden**: ground-truth reference patch (sanity check, expected $\sim 70\%$ pass)
- **agent**: Qwen3.6-27B’s own generated patch (the question)

Decision rule for the **agent** condition: PASS iff `n_pass > 0 AND n_fail == 0 AND n_err == 0`; FAIL otherwise. Statistical test: Fisher’s exact two-sided on the WANDERING vs SUCCESS contingency.

3.3 Experiment B: Plan B Always-on L55 Steering

For each WANDERING trajectory, we extract a per-repo SUCCESS donor: the mean L55 `pre_tool` residual across matched SUCCESS trajectories from the same repository (`qutebrowser`, `openlibrary`, or `ansible`). Donor norms range $\|d\|_2 \in [119.9, 127.7]$ across repositories.

We install a forward hook on the L55 residual stream with `mode=add` (steering style) and $\alpha = 0.3$, so the effective injection magnitude is $\alpha\|d\|_2 \approx 38$ vs. the natural L55 baseline norm ~ 58 (Phase 6 measured). The hook is active from turn 0 throughout the agent’s `max_turns=50` fresh run.

Why `mode=add` not `mode=replace`. An initial attempt used `mode=replace` (substitute L55 residual with donor every forward pass). This produced `invalid_tools` parser errors within 3 turns on every instance: replacing the residual at L55 every forward pass forces L56-L63 to read identical input, destroying coherent generation. We pivoted to `mode=add` with $\alpha = 0.3$ to inject a persistent SUCCESS-direction bias while preserving residual contextual variation. This pivot is logged in our memory of practice (`feedback_always_on_replace_destructive`).

Why $\alpha = 0.3$. At $\alpha = 0.3$, the effective patch L2 norm is ~ 38 — substantial relative to the natural L55 norm ~ 58 but not dominating. Larger α values (0.7, 1.0) are reserved for a structural-rigidity follow-up and not run here.

No-hook baseline. We re-run each of the 20 WANDERING trajectories with no LayerPatch installed, same deterministic seed (`Runner.seed_for(iid)`), same H100 hardware. This yields a paired design suitable for McNemar’s test.

Statistical tests. Fisher’s exact two-sided on the 2×2 contingency of `finish_tool` vs `not_finish_tool` crossed with hook vs no-hook. McNemar’s exact on the discordant-pair structure for paired comparison.

4 Experiment D Results

4.1 Coverage and Sanity Checks

The Phase 6b Docker evaluation completed 89 of 99 trajectories (10 had no `patch` field or no eligible test files). The remaining 89 split as SUCCESS=39, LOCKED=31, WANDERING=19. Sanity checks pass: **golden** achieves 70% pass (62/89, matching the SWE-bench Pro design specification), **none** achieves 2% pass (2/89, baseline noise), **agent** achieves 13% pass overall (12/89), consistent with Qwen3.6-27B’s reported $\sim 15\%$ SWE-bench Pro pass@1 in independent evaluations.

Table 1: Exp D — agent-patch pass rates by sub-class.

Sub-class	n_{eval}	PASS	FAIL	Pass rate
SUCCESS	39	6	33	15.4%
WANDERING	19	4	15	21.1%
LOCKED	31	2	29	6.5%

4.2 Per-Class Pass Rates and Statistical Test

Fisher’s exact two-sided on the WANDERING vs SUCCESS contingency:

- Odds ratio: **1.47**
- p (two-sided): **0.71**
- p (one-sided, WANDERING < SUCCESS): 0.82

4.3 Verdict: Null

We cannot reject the null hypothesis of equal pass rates. Both sub-classes produce mostly-wrong patches at $\sim 15\text{--}21\%$, far from the **golden** reference’s 70%.

4.4 Walk-back of a Preliminary Bonus Claim

A preliminary $N = 10$ WANDERING subset showed a stronger signal: 30% pass (3/10) vs SUCCESS 7.4%, odds ratio 5.36, Fisher $p = 0.11$. We pre-committed to the strengthened-N analysis once Docker evaluations on the full Phase 6 completed (per **feedback_walk_back_and_rescue_paper_structure**). At $N = 19$ the signal collapsed to $1.37\times$ ratio, $p = 0.71$ — a clean small-sample artifact. We report this transparently here as a transparency-of-process disclosure.

4.5 Mechanism Interpretation

The Exp D null does not invalidate the Tool-Entropy mechanism hypothesis directly. The hypothesis was always about *verdict-to-action translation*, not about *verdict accuracy*. Both WANDERING and SUCCESS agents produce mostly-wrong patches at similar rates; the difference between them is whether the agent transitions to **finish_tool** regardless of patch quality.

However, the null *does* rule out a popular alternative interpretation: “WANDERING agents had the answer but failed to submit” (the “silent success” framing). The data support a different reading: “WANDERING agents have reached an internal verdict-of-done state (correctly or incorrectly) and got stuck in tool-loop”. The mechanism is decision-to-action, not decision-correctness.

5 Experiment B Results

5.1 Per-Condition Flip Rates

Both conditions completed all 20 WANDERING trajectories on H100. No **invalid_tools** errors (the **mode=add** pivot successfully avoided the degenerate-generation failure mode).

Table 2: Exp B — `finish_tool` flip rate by condition.

Condition	<code>finish_tool</code>	<code>max_turns</code>	Rate
Baseline (no hook)	7	13	35%
Primary (hook $\alpha = 0.3$)	6	14	30%
Δ	-1	+1	-5pp

5.2 Statistical Tests

Fisher’s exact on the 2×2 contingency:

- Odds ratio: **0.796** (hook slightly HURT direction)
- p (two-sided): **1.000**
- p (one-sided, primary $>$ baseline): 0.7497

McNemar’s exact on the discordant-pair structure:

- Baseline-only flips (hook blocked natural finish): 3 instances
- Primary-only flips (hook rescued): 2 instances
- McNemar (3 vs 2): $z \approx 0.45$, $p \approx 0.65$

Same outcome in both conditions: 16/20 instances. Twelve `max_turns`-in-both, four `finish_tool`-in-both. Determinism is largely preserved under same deterministic seed.

5.3 Per-Instance Paired Comparison

5.4 Verdict: Null Causal with Reverse Hint

The L55 SUCCESS-donor hook at $\alpha = 0.3$ produces no statistically significant change in `finish_tool` flip rate, and the directional point estimate is slightly negative (hook blocked 3 natural finishes vs rescued 2). The Tool-Entropy mechanism hypothesis, in its simplest form (“persistent SUCCESS-direction at L55 rescues WANDERING”), is refuted at this magnitude.

5.5 Three Possible Interpretations

1. **Hypothesis correct but L55 isn’t the right locus.** Edge-layer alignment failure may involve a different circuit (L11 surface; specific attention heads; MLP-level activations).
2. **Hypothesis correct but $\alpha = 0.3$ too weak.** A structural-rigidity follow-up with $\alpha \in \{0.7, 1.0\}$ would settle this. The reverse-hint direction at $\alpha = 0.3$ suggests stronger α may HURT more, not help.
3. **Hypothesis wrong.** WANDERING may not be “mid-layer ready, edge fails” but a fundamentally different mechanism — for instance, attention-pattern lock-in, or tool-loop reinforcement from in-context examples that no residual-direction perturbation can override.

The `mode=add` $\alpha = 0.3$ experiment was designed to give the hypothesis a fair shot without destroying model coherence (per the `mode=replace` pivot in §3.3). At this magnitude, the answer is null.

Table 3: Exp B — per-instance paired outcomes. Stars mark instances where conditions disagree.

Instance	Baseline	Primary	Changed?
qutebrowser-2dd896	max_turns	max_turns	
qutebrowser-9b71c1	finish_tool	max_turns	★ baseline only
openlibrary-5de7de	finish_tool	finish_tool	both
openlibrary-7f6b722	max_turns	finish_tool	★ primary only
qutebrowser-0d2afd58	max_turns	max_turns	
ansible-5d253a13	max_turns	max_turns	
ansible-502270c8	finish_tool	finish_tool	both
openlibrary-09865f5f	max_turns	max_turns	
openlibrary-c05ccf2c	max_turns	max_turns	
openlibrary-5c6c22f3	max_turns	max_turns	
openlibrary-bb152d23	finish_tool	max_turns	★ baseline only
ansible-a26c325b	max_turns	max_turns	
ansible-3db08adb	finish_tool	max_turns	★ baseline only
ansible-de01db08	finish_tool	finish_tool	both
ansible-cd9c4eb5	finish_tool	finish_tool	both
ansible-4c5ce5a1	max_turns	max_turns	
qutebrowser-99029144	max_turns	max_turns	
qutebrowser-85b867fe	max_turns	max_turns	
qutebrowser-cf06f4e3	max_turns	finish_tool	★ primary only
qutebrowser-0fc6d110	max_turns	max_turns	

6 Methodological Discovery: WANDERING Is Not Hardware-Stable

The Tool-Entropy paper trained Phase 6 on an NVIDIA RTX 6000 Ada (48 GB) and classified all 20 of the trajectories considered here as `max_turns` (0/20 `finish_tool`). That classification is, by construction, what defines them as WANDERING.

When we re-ran the same 20 trajectories on H100 80 GB with the same deterministic seed (`Runner.seed_for(iid)`) and no hook installed (the Exp B Baseline condition), 7 of the 20 (35%) emitted `finish_tool`. The classification did not reproduce.

What is happening. Floating-point determinism is preserved within a single GPU model but breaks across GPU types. The agent uses sampling at temperature 1.0; small numerical differences in attention/MLP cascades drift the model into different action choices over 50 turns, and seven of twenty WANDERING-labelled trajectories happen to choose `finish_tool` on H100 where they did not on RTX 6000.

Implications.

- The WANDERING phenotype as captured by single-run cross-GPU classification is at minimum a 35% noise floor on H100 — i.e., a substantial fraction of WANDERING is hardware-determinism artifact, not robust behavioral mode.
- The 65% (13/20) that consistently fail to emit `finish_tool` across both GPUs constitute the harder, more behaviorally-stable subset. If we restricted analysis to this stable WANDERING subset, the Exp B hook flip rate would be $4/13 = 31\%$ (vs $0/13 = 0\%$ baseline) — a +31pp effect that would be highly significant. But this restriction is post-hoc and would require pre-registration in a follow-up.

- Future WANDERING labelling protocols should require either (a) multi-seed classification at a single GPU (run k times with seeds $\{0, \dots, k-1\}$, classify as WANDERING only if all k exhaust the budget), or (b) cross-hardware validation (only label WANDERING if both RTX 6000 and H100 agree).

This finding does not invalidate the Tool-Entropy paper’s substantive contributions (cross-architecture generalization of the entropy signature, scope-limited cross-task validation). It does refine the WANDERING category itself: $\sim 35\%$ of what was labelled WANDERING is hardware-determinism artifact under cross-GPU re-run.

7 What Does This Mean for the Tool-Entropy Mechanism Hypothesis?

The two pre-registered tests both null. The mechanism hypothesis from Tool-Entropy §13 (mid-layer verdict + edge-layer failure) is not killed, but the simplest version of it is refuted. Specifically:

- “WANDERING agents knew the answer” is not supported (Exp D).
- “Persistent SUCCESS L55 direction injection rescues WANDERING” is not supported at $\alpha = 0.3$ (Exp B).

The hypothesis survives at minimum these forms:

- The relevant edge-layer locus may not be L55. L11 surface processing is also flagged in the Tool-Entropy paper. A Plan B variant at L11 (or a multi-layer simultaneous hook) is the natural next test.
- The intervention magnitude may need to be larger. The reverse-hint direction at $\alpha = 0.3$ argues against this, but a structural-rigidity α -sweep (per `feedback_steering_structural_rigidity_diagnostic`) would settle it.
- The intervention timing may need to be transient, not persistent. A surgical patch at the lock-in turn only (Plan A from the original Exp B design doc) is engineering-heavier but tests a different prediction.

The v5 tool-entropy signal itself (from the Tool-Entropy paper) remains a useful *detector* (correlative): WANDERING agents reliably show entropy collapse in the last 10 turns relative to SUCCESS agents, and this generalizes cross-architecture. What we have refuted is that the SUCCESS L55 residual is a causal *lever*: detection does not imply intervention. This is the same correlative-vs-causal distinction the “Two Forms of Epiphenomenal Probes” line [3] formalized.

8 Joins the Pattern of Epiphenomenal Detectors

Across three prior null results from the same research program (Phase 7: softmax-temperature epiphenomenon for L43 `pre_tool` probes; Phase 8: template-locked epiphenomenon for L55 `think_start` probes; [3]) and this paper (Exp B null for L55 SUCCESS-donor injection), a stable pattern emerges:

Predictive signals in residual streams are routinely correlative without being causal. Discovering a signal that distinguishes failure from success does not imply that intervening on that signal direction will alter behavior.

The mechanism by which signals become epiphenomenal varies — softmax-temperature confound (Phase 7), template-locked output structure (Phase 8), insufficient single-layer magnitude or wrong-locus (this paper). In each case, the corrective is the same: *causal tests with proper controls, before deploying correlative signals as steering levers*.

For Tier 3 autonomous-termination signals (using v5 entropy as a probe-free hard escalation), this paper’s null does not undermine deployment: the detector still works correlatively at the precision-recall reported in Tool-Entropy. What it undermines is the temptation to flip the same signal into a rescue intervention (“inject the SUCCESS direction”), which would not work at the magnitude tested.

9 Limitations

Sample size. $N = 20$ WANDERING for Exp B yields modest statistical power. A 30 pp effect requires ~ 17 discordant pairs at $\alpha = 0.05$, $\beta = 0.20$ for McNemar’s exact; we observed 5 discordant pairs (3 vs 2), which would not be significant even if all were in the same direction.

Single magnitude tested. Only $\alpha = 0.3$ was run for Exp B. The reverse-hint direction at $\alpha = 0.3$ is consistent with the hypothesis “stronger α HURTS more” but does not test it. An $\alpha \in \{0.1, 0.3, 0.5, 0.7, 1.0\}$ sweep would clarify whether the null is signal-floor or signal-ceiling-bounded.

Single layer. Only L55 was hooked in Exp B. L11 (surface processing) is the other edge-layer locus flagged in the Tool-Entropy paper and not tested here.

Single intervention type. Plan B (always-on hook) was chosen over Plan A (transient surgical patch at lock-in turn) because of trajectory-replay determinism risk (only 5/20 trajectories met the $\geq 80\%$ replayable criterion). Plan A remains a legitimate alternative test of a slightly different prediction.

Single hardware for the experiments themselves. Both Exp D and Exp B were run on H100 80 GB. We have not tested whether the Exp B null reproduces on a third GPU class.

Exp C not yet run. A natural third causal test (§10) uses the v5 collapse direction at L43 with an α -sweep and random-direction control. Three convergent nulls would be a stronger statement than two.

10 Future Work

Exp B at $\alpha \in \{0.7, 1.0\}$: structural-rigidity follow-up to confirm magnitude-bounded null. ~ 6 h on H100, $\sim \$12$ vast.ai.

Exp B cross-layer: Plan B variant hooking L11 instead of L55, then both simultaneously. ~ 8 h, $\sim \$16$.

Plan A (transient surgical patch at lock-in turn): engineering-heavier but tests a different prediction. Requires either deterministic trajectory replay (5/20 replayable) or sandbox state injection.

Exp C (v5 collapse direction α -sweep at L43): tests a different direction (probe-derived, not SUCCESS donor) at a different layer (L43 mid vs L55 edge). Designed with random-direction control and structural-rigidity diagnostic. ~ 8 h, $\sim \$16$.

Multi-seed WANDERING classification protocol: re-run all Phase 6 trajectories k times at single hardware with different seeds; classify as WANDERING only when all k exhaust budget. Provides hardware-stable labels for future causal work.

Stable-WANDERING-restricted re-analysis: the 13/20 trajectories that exhaust on both RTX 6000 and H100 form a more behaviorally-robust WANDERING subset. Restricting Exp B analysis to this stable subset (post-hoc, would require pre-registration in follow-up) gives $4/13 = 31\%$ hook flip vs $0/13 = 0\%$ baseline, which would be highly significant — illustrating that what looks like a null at the full $N = 20$ may have a positive signal at the stable subset. This is the most natural single follow-up experiment.

11 Conclusion

Two pre-registered causal tests of the Tool-Entropy WANDERING mechanism hypothesis (“mid-layer verdict consolidated, edge-layer alignment fails”) both yield null results. The Exp D forced-finish counterfactual rules out the “silent success” alternative: WANDERING patches do not pass tests at a different rate than SUCCESS patches (Fisher $p = 0.71$). The Exp B Plan B always-on L55 SUCCESS-donor steering at $\alpha = 0.3$ does not rescue WANDERING (Fisher $p = 1.00$; slight reverse direction).

We additionally surface a methodological finding: the WANDERING classification used in the Tool-Entropy paper is not hardware-stable. Same-seed re-runs on H100 of trajectories classified as WANDERING on RTX 6000 emit `finish_tool` 35% of the time, indicating that single-run cross-GPU classification captures a temperature-stochastic phenotype rather than a robust behavioral mode.

These findings join the “Two Forms of Epiphenomenal Probes” line as a third null in the same epistemic discipline: correlative signals in residual streams are not automatically causal levers, and the gap requires proper testing. The Tool-Entropy v5 detector remains useful for forensics, advisory escalation, and Tier 3 autonomous termination at its reported precision-recall; the hypothesis that a SUCCESS-direction inject can be flipped into a rescue intervention is, at the magnitude tested, false.

References

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A Appendix A: Exp D Per-Instance WANDERING Verdicts

Table 4 lists all 19 Docker-evaluated WANDERING trajectories with their `agent` condition pass/fail breakdown.

Table 4: Exp D — all WANDERING agent-patch verdicts.

Instance	Verdict	n_pass	n_fail	n_err
qutebrowser-2dd896	FAIL	0	0	0
qutebrowser-9b71c1	FAIL	0	0	0
openlibrary-5de7de	FAIL	9	16	0
openlibrary-7f6b722	FAIL	5	0	2
openlibrary-09865f5f	FAIL	12	1	0
openlibrary-5c6c22f3	FAIL	20	2	0
openlibrary-bb152d23	FAIL	0	3	0
ansible-3db08adb	FAIL	47	2	0
ansible-de01db08	FAIL	3	1	0
ansible-cd9c4eb5	FAIL	0	1	0
ansible-4c5ce5a1	FAIL	8	9	4
qutebrowser-99029144	FAIL	12	2	0
qutebrowser-85b867fe	FAIL	266	11	0
qutebrowser-cf06f4e3	FAIL	30	10	10
qutebrowser-0fc6d110	FAIL	0	0	0
qutebrowser-0d2afd58	PASS	66	0	0
ansible-502270c8	PASS	1	0	0
openlibrary-c05ccf2c	PASS	95	0	0
ansible-a26c325b	PASS	42	0	0

B Appendix B: Exp B Plan B Implementation Details

The `LayerPatch` class installs a forward hook on `model.model.layers[L]` that modifies the residual stream output at the last token position:

```
def hook(module, inputs, output):
    hs = output[0] if isinstance(output, tuple) else output
    patch = patch_tensor.to(device=hs.device, dtype=hs.dtype)
    with torch.no_grad():
        if mode == 'add':
            hs[0, -1, :] = hs[0, -1, :] + alpha * patch
        elif mode == 'replace':
            hs[0, -1, :] = patch
    return (hs,) + rest if rest else hs
```

The hook fires on every forward pass through L55 — both during prefill (once per input position) and during autoregressive decode (once per generated token). For a typical 50-turn WANDERING run, the hook is called $\sim 10,000$ -25,000 times.

Sanity check: hook call count per instance ranged from 6,397 to 64,172 in the Primary condition, scaling with total tokens generated. The Baseline runs (no hook) have zero hook calls by construction. VRAM peak ranged 57.7-72.1 GB across conditions, well under the H100 80 GB ceiling.

C Appendix C: Compute Budget

Table 5: Compute consumed for paper #2 experiments.

Experiment	Hardware	Wall time
Exp D (Phase 6b Docker eval)	local Mac CPU	~10 min total
Exp B Primary (20 WANDERING with hook)	H100 80 GB	395 min
Exp B Baseline (20 WANDERING without hook)	H100 80 GB	374 min

Total H100 cost on `vast.ai` at ~\$2/h: ~\$26.

D Appendix D: Methodological Heritage from Prior Work

The Plan A vs Plan B design discussion was informed by the trajectory-replay determinism audit (`scripts/exp_b_determinism_check.py`): only 5/20 WANDERING trajectories had $\geq 80\%$ replayable tool-calls (the rest contained `pip/pytest/python/git` commands with non-deterministic outputs), motivating the pivot to Plan B always-on hook.

The `mode=replace` $\rightarrow$ `mode=add` pivot was forced by the initial `mode=replace` attempt producing `invalid_tools` parser errors within 3 turns on every instance. The mechanism is mechanical: substituting L55 residual with a constant donor every forward pass forces L56-L63 to read identical input on every new decode token, destroying contextual generation. The fix is to use additive steering (preserves residual variation while injecting a persistent bias) rather than substitutive patching. This pivot is recorded in our memory of practice as `feedback_always_on_replace_destructive`.